1.) Given,

$G_x = 24, \quad G_y = 18$

1. Gradient Magnitude

the gradient magnitude is

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900} = 30$$

2. Gradient Direction

The gradient direction is

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$= \tan^{-1}(0.75)$$

$$= 36.87^\circ$$

Gradient Magnitude = 30

Gradient direction = $36.87^\circ$

2. Given gradient magnitudes

| Pixel | P1 | P2 | P3 | P4 | P5 |
|---|---|---|---|---|---|
| Magnitude | 25 | 50 | 90 | 60 | 30 |

Non-Maximum Suppression

- P1 : Boundary pixel → Cannot compare both sides
  Suppressed.
- P2 : 50 : Compare with P1 = 25 & P3 = 90.
  Since 50 < 90, Supeess P2
- P3 - 90 : Compare with P2 = 5 and P4 = 60
  Since 90 > 50 and 90 > 60, keep P3
- P4 = 60 : Compare with P3 = 90 and P5 = 30.
  Since 60 < 90, Supeess P4
- P5 = Boundary pixel → ~~Suppressed.~~

0, 0, 90, 0, 0

∴ P3 = 90 is the only pixel retained.

3) Given:
   High threshold = 100
   Low threshold = 50

Gradient Magnitudes

| Pixel | P1 | P2 | P3 | P4 | P5 |
|---|---|---|---|---|---|
| Magnitude | 135 | 90 | 45 | 120 | 70 |

# Classification Rules in Canny Edge Detection

- If Magnitude ≥100 → Strong Edge
- If Magnitude b/w 50 & 100 → Weak Edge
- If Magnitude < 50 → Non-edge

Classification

- P1 = 135 → Strong Edge
- P2 → 90 → weak Edge
- P3 = 45 → Non-edge
- P4 → 120 → strong edge
- P5 → 70 → weak edge
- P6 → 20 → Non-edge

Strong Edges = P1, P4

weak edges = P2, P5

Non edges = P3, P6

4) Given

- High threshold = 100
- Low threshold = 50

| Pixel | P1 | P2 | P3 | P4 | P5 |
|---|---|---|---|---|---|
| Magnitude | 130 | 95 | 40 | 115 | 65 |

1. Strong Edges

Magnitude $\geq 100$

$P1 = 130$

$P4 = 115$

$P7 = 105$

$\boxed{\text{Strong Edges} = P1, P4, P7}$

2. Weak Edges

$50 \leq \text{Magnitude} < 100$

$P2 = 95$

$P5 = 65$

$\boxed{\text{Weak Edges} = P2, P5}$

3. Non - Edges

Magnitude $< 50$

$P3 = 40$

$P6 = 25$

$\boxed{\text{Non- Edges} = P3, P6}$

4. Final Edge Pixels After Hysteresis

$P2 \rightarrow P1$ = retained

P5 not connected to strong edge = Removed

Strong edges P1, P4, P7 → Retained.

∴ Final Edge pixels = P1, P2, P4, P7

1) Given,

- Edge points $P_1 = (5, 4)$, $P_2 = (7, 6)$
- Candiate Centers $C_1 = (2, 4)$, $C_2 = (4, 3)$

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

1. Center $C_1 = (2, 4)$

For $P_1 = (5, 4)$

$$r = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3$$

For $P_2 = (7, 6)$

$$r = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{29} = 5.39$$

$C_1 = r_1 = 3, 5.39$

$\therefore$ No Matches

2. Center $C_2 = (4, 3)$

For $P_1 = (5, 4)$

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$$

For $P_2 = (7, 6)$

$$r = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$$

$C_2 = r = 1.41, 4.24$.

$\therefore$ No Matches

2) Given,

Edge point $P=(8,6)$

Candiate Center $C=(4,3)$

$$r=\sqrt{(x-h)^2+(y-k)^2}$$

$$=\sqrt{(8-4)^2+(6-3)^2}$$

$$=\sqrt{4^2+3^2}$$

$$=\sqrt{16+9}=5$$

3) Given,

Center $C=(5,4)$

Edge point $P=(8,8)$

$$r=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$=\sqrt{(8-5)^2+(8-4)^2}$$

$$=\sqrt{3^2+4^2}=\sqrt{9+16}$$

$$=5$$